

Practical: ARP Spoofing Attack Using Ettercap

Aim

To demonstrate an ARP spoofing attack using Ettercap in a controlled environment and understand how attackers can intercept network traffic by manipulating the Address Resolution Protocol.

Objectives

- To understand the working of ARP and ARP spoofing attacks.
- To simulate a Man-in-the-Middle attack using Ettercap.
- To analyze intercepted traffic using Wireshark.
- To observe how ARP poisoning affects communication between network devices.

Requirements

- Victim Machine: System running Ubuntu
- Attacker Machine: System running Kali Linux
- Tools Required: Ettercap, Wireshark, nmap

All machines must be connected to the same network.

ARP Spoofing Man-in-the-Middle



Step 1: Identify IP Addresses of Machines

On the Victim Machine

- Open the terminal and type:
ip a
- Note the IP address and MAC address of the victim machine.

On the Attacker Machine (Kali)

- Open terminal and type:
ip a
- Note the attacker's IP and MAC address.

The MAC address is important because ARP maps IP addresses to MAC addresses.

Step 2: Identify the Network Gateway

- The gateway is usually the router of the network.
- Run the command on Kali:

ip route

- Example output:
default via 192.168.1.1
So the gateway IP = 192.168.1.1
- To find the gateway MAC address:

arp -n

Example output:

Address	HWaddress
192.168.1.1	00:50:56:C0:00:08

- Record this MAC address because **during spoofing this mapping will change.**

Step 3: Scan the Network for Active Hosts

- Use nmap to identify devices on the network.

nmap -sn 192.168.1.0/24 (This is just an example. Use your own network range)

- This will display all active machines on the network.
- Identify:
Victim IP
Gateway IP

Step 4: Enable IP Forwarding

- Before performing ARP spoofing, enable packet forwarding so traffic continues flowing through the attacker machine.

- Run on Kali:
`echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward`
- This allows the attacker system to forward packets between victim and gateway.

Step 5: Launch Ettercap

- Start Ettercap in text mode:
`sudo ettercap -T -i eth0 -M arp:remote /Victim_IP/ /Gateway_IP/`
- Note: You will select your network interface (eg. eth0 or enp0s1 etc.)
- Example:
`sudo ettercap -T -i eth0 -M arp:remote /192.168.1.20// /192.168.1.1//`
- Meaning of parameters:
 - -T: Text mode
 - -i eth0: Network interface
 - -M arp:remote: remote Perform ARP spoofing
 - /Victim_IP/ : Target victim
 - /Gateway_IP/ : Target router
- During the attack, the attacker sends fake ARP replies to both the victim and the gateway.

Step 6: Monitor Network Traffic

- While the attack is running, open Wireshark on the attacker machine.
- Start capturing packets on the network interface being used (e.g., eth0 or wlan0).
- You may observe ARP replies being sent repeatedly, indicating ARP poisoning.
- Apply the following display filter:
`arp`
- You should observe multiple ARP Reply packets, such as:
ARP Reply: 192.168.1.1 is at 08:00:27:11:22:33
- Important observation:

IP Address	MAC Address
Gateway IP	Attacker MAC
- This indicates the attacker is pretending to be the gateway.
- In the Protocol Details pane, Wireshark may display warnings such as:
Duplicate IP address detected
This occurs because two devices appear to claim the same IP address, which is a typical indicator of ARP poisoning.

Step 7: Observe Intercepted Network Traffic

- Once ARP spoofing is active, the attacker becomes the Man-in-the-Middle.
- Traffic flow changes to:
Victim → Attacker → Gateway
This means the attacker can capture packets passing between the victim and gateway.
- On the victim machine, open a browser and visit:

http://testphp.vulnweb.com/login.php

- Log in with the username “test” and password “test”.
- Also update the other details such as name, address, phone, credit card etc., after logging in and select update.
- Since this site uses unencrypted HTTP, its traffic can be captured.
- In Wireshark, apply filter and analyze the protocol details pane:

ip.addr == Victim_IP

http.request.method == "POST"

You should see packets such as:

- DNS queries between the victim and the gateway, responses from the gateway, HTTP GET requests, Website URLs, Form submissions, etc.
- This confirms traffic is passing through the attacker.
- Analyse the packets to identify the unencrypted username, password, and other details entered on the website earlier.

Step 8: Verify ARP Spoofing on Victim Machine

- On the victim machine, check the ARP table:

arp -n

- You may notice that the gateway IP now maps to the attacker’s MAC address, which indicates ARP spoofing.

Step 8: Stop the Attack

- Terminate the attack by pressing:

q (on the keyboard)

in the Ettercap terminal.

Conclusion

This experiment demonstrated how ARP spoofing can be used to perform a Man-in-the-Middle attack. By sending forged ARP replies, the attacker manipulated the IP-to-MAC address mapping between the victim and gateway. This allowed the attacker to intercept network traffic and observe unencrypted communications. Packet analysis in Wireshark revealed repeated ARP replies, duplicate IP warnings, and captured network traffic, illustrating the security weaknesses of the Address Resolution Protocol.